Base-field theta setup gate

KUMMER-THETA-AFFINE-20261022

Models: 21; with Fp theta tuple: 21.

Each model: 20 ordered finite-root pairs, 10 square pairs,

60 eligible Rosenhain root orders; fixed gauges b=f=1.

Qualifying tuples: 1680; counts per model: 48, 80 or 112.

Independent lookup replay matches every model's count and checks

each candidate against the original equations by cross product.

First p=739 selection: (lambda,mu,nu)=(296,16,10);

(a,b,c,d,e,f)=(369,1,255,605,719,1).

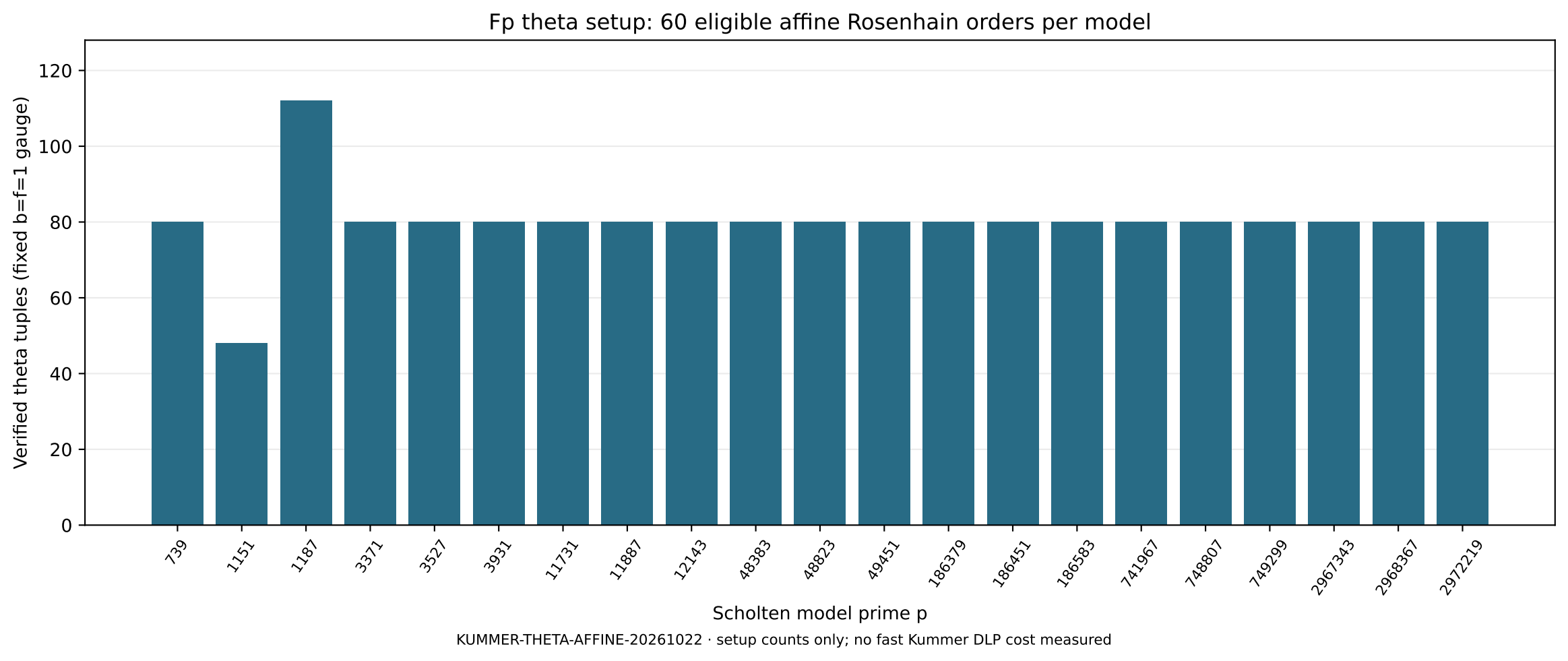
Source: Chung, Costello, Smith, section 7.1:

https://www.microsoft.com/en-us/research/wp-content/uploads/2016/02/recovery.pdf

Boundary: affine root orders and Fp constants only.

No xADD/xDBL, signed DLP collector, operation cost or rho ratio measured.

Raw: results/theta-models.jsonl; definition: DATA_DICTIONARY.md.


Fp theta setup: 60 eligible affine Rosenhain orders per model
Verified theta tuples (fixed b=f=1 gauge)
0
20
40
60
80
100
120
739
1151
1187
3371
3527
3931
11731
11887
12143
48383
48823
49451
186379
186451
186583
741967
748807
749299
2967343
2968367
2972219
Scholten model prime p
KUMMER-THETA-AFFINE-20261022 · setup counts only; no fast Kummer DLP cost measured